

13. the current sky panel shows the "double helix" fibers, which are essentially word lines - time-like latitudes on the holographic shell. What we are looking for next is to construct the complementary space-like set of fibers - the meridians of the holographic shell retraced in the 3D space-time image of the physical sky. Please investigate the properties of such objects and provide a detailed proposal of the construction and physical interpretation

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The complementary spatial family is well defined, but it should not be produced by merely rotating the current double helix. The canonical construction is an exact transpose of the 156-event affine lattice: fixed spatial address gives a time-like history, while fixed source chronon gives a 13-cell space-like meridian.

One important refinement is necessary. The current helix belongs to the temporal sector, but a frozen helix is not literally one worldline: it stitches together retarded events with different source chronons. The new mode must regroup those events by common source chronon.

1. Exact finite construction

Let

$$U_a(y) = y + a, \quad D_m(y) = g^m y,$$

where $a \in \mathbb{F}_{13}$, $m \in C_{12}$, and $g = 2$. Write an affine event as

$$E_{m,a} = D_m U_a.$$

Its shell registration is

$$R_m(a) = a g^m.$$

The 156-element Borel event set

$$B = AGL(1, 13), \quad |B| = 13 \cdot 12 = 156$$

then has two transverse decompositions:

Family	Fixed coordinate	Varying coordinate	Leaves
Time leaf W_a	a	$m \in C_{12}$	13 leaves of 12 events
Space meridian Σ_m	m	$a \in \mathbb{F}_{13}$	12 leaves of 13 events

Explicitly,

$$W_a = \{E_{m,a} : m \in C_{12}\},$$

$$\Sigma_m = \{E_{m,a} : a \in \mathbb{F}_{13}\}.$$

Every pair intersects exactly once:

$$W_a \cap \Sigma_m = \{E_{m,a}\}.$$

This is the exact meaning of temporal/spatial complementarity. It does not depend on a Euclidean right angle in the rendered chart.

At fixed m ,

$$R_m(a+1) - R_m(a) = g^m.$$

Thus every step along a meridian is the same additive displacement. That is the finite, metric-free meaning of a “straight” spatial line.

Evolution transports the spatial direction according to

$$D_m U_a D_m^{-1} = U_{g^m a},$$

and

$$R_{m+1}(a) = g R_m(a).$$

So one chronon advances the entire spatial section coherently. Also,

$$R_{m+6}(a) = R_m(-a),$$

meaning that there are 12 oriented meridians but six unoriented shell great circles.

This interpretation agrees with the FRC papers’ identification of additive motion with meridians, multiplicative drive with latitudes, and Dirac development with the meridional direction. The papers explicitly treat these physical names as tagged FRC interpretations. [Schrödinger and Dirac Dynamics over Finite Substrate](#) , [Quantum Observation over Finite Relational Substrate](#)

2. What the current sky actually displays

For a current sky node (s, ϵ, k) , the implementation uses

$$a = \epsilon k \pmod{13},$$

$$x_s = a g^{\tau-k} M_s,$$

with

$$M_s = (3, 9, 1) = (g^4, g^8, g^0).$$

Define

$$\mu_s = (4, 8, 0).$$

Then every production node has the exact meridian address

$$m = \tau - k + \mu_s \pmod{12}$$

and label

$$x_s = a g^m.$$

At fixed display time τ , varying k therefore varies m . The visible helix is a retarded/null diagonal through the time histories, rather than an equal- m spatial section.

A literal worldline holds (a, s) fixed and varies τ ; its colatitude is fixed and it traces a time-like latitude. Your characterization of the existing mode as belonging to the time sector is therefore right, but the single displayed loop is more precisely a causal stitching of those histories.

This follows directly from the current implementation and verifier formulas. I evaluated `main` at commit `c76348b`. [Current simulation](#) , [source at the evaluated commit](#) , [sky verifier](#)

3. Reconstructing one complete meridian

The strongest construction is to transpose the existing verified event allocation.

For a target m :

1. Compute the 72 production nodes at τ .
2. Compute the complementary 72 nodes at $\tau + 2$, or $\tau - 2$ in a causal replay.
3. Assign every node its exact address

$$(a, m) = (\epsilon k, \tau - k + \mu_s).$$

4. Select the unique node for each $a = 1, \dots, 12$ having the chosen m .
5. Add the origin event $(0, m)$.
6. Connect the stations in additive order

$$0, 1, 2, 3, 4, 5, 6, -6, -5, -4, -3, -2, -1, 0.$$

For every meridian:

$$6 \text{ nodes at } \tau + 6 \text{ nodes at } \tau + 2 + 1 \text{ origin} = 13.$$

Across all meridians,

$$12(6 + 6 + 1) = 72 + 72 + 12 = 156.$$

The transposed selection has useful exact balances:

- four nonzero nodes on each of the three radial shells;
- three nodes in each sector:
direct@ τ , echo@ τ , direct@ $\tau + 2$, echo@ $\tau + 2$;
- one and only one representative of every nonzero additive cell.

This thread crossing radial shells and route sectors is the primary physical-sky meridian. It uses the existing totality rather than introducing a new geometric degree of freedom.

4. Direct historical reconstruction

The same result has a clear causal interpretation.

For centered cell $a \in \{-6, \dots, 6\}$, set

$$d = |a|, \quad \epsilon = \text{sgn}(a).$$

A cell belonging to source section m arrives by the direct route at

$$T = m + d.$$

The current label law then gives

$$a g^{T-d} = a g^m.$$

Its echo uses

$$d^\vee = 13 - d, \quad \epsilon^\vee = -\epsilon, \quad T^\vee = m + d^\vee,$$

and produces the same cell and value:

$$-\epsilon(13 - d) \equiv \epsilon d \pmod{13},$$

$$a g^{T^\vee - d^\vee} = a g^m.$$

Thus the direct and echo points are two observational lifts of one base spatial event. Their distances sum to 13 and their colatitudes sum to π .

Operationally:

current sky $\mathcal{N}_\tau$ = events arriving at one reception chronon,

whereas

Σ_m = events reconstructed at one common source chronon.

The meridian is therefore the Subject's synchronized spatial section reconstructed from its retarded sky.

For a strictly accumulated history, the latest fully available direct section is $m = T - 6$; all nonzero echoes require $m \leq T - 12$. The $k = 13$ self-echo is the formal null closure.

5. The C_{26} spatial lift

The base meridian is C_{13} . Its oriented sky route is the double cover C_{26} :

- $j = 0$: observer/null anchor;
- $j = 1, \dots, 12$: first circuit;
- $j = 13$: antipodal null self-echo;
- $j = 14, \dots, 25$: return sheet;
- $j = 26$: closure.

The temporal and spatial lifts share exactly one sign:

$$\widetilde{D}^{12} = -I, \quad \widetilde{U}^{13} = -I,$$

with

$$\text{ord}(\widetilde{D}) = 24, \quad \text{ord}(\widetilde{U}) = 26.$$

Equivalently, the lifted event-address set is

$$\widetilde{E} = (C_{24} \times C_{26}) / \langle (12, 13) \rangle, \quad |\widetilde{E}| = 312.$$

It contains:

- 13 temporal fibers of length 24;
- 12 spatial fibers of length 26;
- two lifted intersections for each time/space pair;
- one base event after removing the common sign.

This C_{312} object should remain an event-address cover. The physical Borel events retain their noncommutative semidirect composition.

6. Geometry in the 3D chart

The current unflowed sky indices are, schematically,

$$\theta = \frac{k\pi}{13}, \quad \alpha = \tau + 3k \pmod{24}, \quad \beta = \epsilon k \pmod{13}.$$

For a fixed source meridian, substitute $\tau = m + k$:

$$\alpha_{\text{arrival}} = m + 4k \pmod{24}.$$

This is the native spacetime arrival trace: one step arises from the later reception time and three from the existing phase transport.

If each arrival is pulled back by k temporal-cover steps into a common Subject frame, then

$$\alpha_{\Sigma} = m + 3k \pmod{24}.$$

This synchronized skeleton can have the same general locus as the current guide. What changes is the event grouping and value law, not merely the stroke shape. That is why a genuine space mode requires reconstructed nodes rather than the current “change fibers only” toggle used between Helix and Hopf.

Optional smooth backbone

A useful faint chart guide follows by freezing the temporal-cover address while retaining the spatial half-angle. Let

$$\theta_j = \frac{\pi j}{13}, \quad \alpha_{\ell} = \frac{\pi \ell}{12}.$$

Up to the current axis signs,

$$\Gamma_{\ell,s}(j) = R_s \begin{pmatrix} \sin \theta_j \cos(\alpha_\ell + \theta_j) \\ \sin \theta_j \sin(\alpha_\ell + \theta_j) \\ \cos \theta_j \end{pmatrix}.$$

In axes rotated by α_ℓ ,

$$u^2 + \left(v - \frac{R_s}{2}\right)^2 = \left(\frac{R_s}{2}\right)^2.$$

Thus the undressed guide is a Viviani-type sphere–cylinder intersection rather than an ordinary planar great circle. It:

- closes after 26 stations;
- passes through the two polar anchors;
- has a continuous figure-eight crossing at $j = 6.5$ and 19.5 ;
- has no registered node at that crossing because $p = 13$ is odd;
- places the crossing exactly between horizon rows 6 and 7.

This is a particularly attractive chart image of the distributed spatial sign. Retardation, the fine C_{232} curl, and the Hopf flow can then dress this faint guide. Exactness must remain attached to the finite stations; the smooth curve is chart interpolation.

7. Topology and terminology

The new objects are not additional Hopf fibers.

A spatial meridian:

- is a 13-event C_{13} cycle;
- has a 26-station route/spin lift;
- passes transversely through 13 different Hopf leaves;
- is an equal-chronon register section, not observer-changing boost motion;
- is not a material trajectory or force line.

In the full event set the 12 meridians are distinct. Their 3D projections share polar/null anchors, so the projected family is best called a:

meridional pencil or singular spatial foliation.

It is not a regular Hopf fibration. Screen crossings between interpolated strokes are chart coincidences unless registered finite addresses coincide.

Sweeping Σ_m through all m produces a two-dimensional spacetime worldsheet. An individual Σ_m is one spatial slice, not the worldsheet itself.

8. Physical interpretation

The construction completes the three finite dynamical directions:

Sector	Exact structure	FRC reading
Time	C_{12} , lifted to C_{24}	winding histories and free drive
Space	C_{13} , lifted to C_{26}	synchronized meridians and kinetic translation
Boost	C_{14}	observer change, Hopf leaf, mass channel

Accordingly:

- time is progression along $W_{a,i}$;
- space is the transverse section $\Sigma_{m,i}$;
- the visible sky is the null/echo diagonal $\mathcal{N}_{\tau,i}$;
- boost motion is vertical displacement along the Hopf leaf.

“Spacelike” here means equal registered chronon. Distance remains the exact number of additive/decoherence steps. No continuum interval is required.

In the current Objectless episode the meridian supplies spatial support only. A later Object may place amplitudes or kinetic evolution on that support; the meridian itself should not be portrayed as a particle path.

9. Recommended rendering

I would add:

Retarded helix | Space meridians | Hopf / boost

For the spatial mode:

1. Show one selected Σ_m as the bright primary thread.
2. Use blue **#3987e5**, matching shell M_0 .
3. Draw Σ_{m+3} optionally in red as the momentum-quarter dual M_3 .
4. Show the six τ stations solid.
5. Show the complementary six $\tau \pm 2$ stations dim or dashed.
6. Add the phase-labelled origin once.
7. Preserve direct/echo styling and the node arrows/tints.
8. Show the other eleven meridians only as a faint atlas or on hover.
9. Optionally show the three fixed-rung C_3 history lifts as faint cover guides.
10. Keep the bright exact thread piecewise: its additive adjacency jumps between radial shells, so a smooth geodesic is not presently derived.

The new mode intentionally uses a different event section. The existing Helix↔Hopf “same nodes, different fibers” invariant should remain scoped to those two modes.

10. Implementation and verification

The source should first refactor `skyState` into a pure historical evaluator such as:

```
JavaScript
skyStateAt({ tS, tC, sigC, view, frac })
```

This is needed because the present function reads global `sigC`, `view`, and interpolation state. Historical nodes must be evaluated with their own arrival states.

A new `verify-meridian.js` should establish at minimum:

- 12 meridians of 13 events and 13 histories of 12;
- singleton $W_a \cap \Sigma_m$;
- $D_m U_a D_m^{-1} = U_{g^m a}$;
- $R_m(a+1) - R_m(a) = g^m$;
- six current plus six complementary nodes per meridian;
- one event per nonzero additive address;
- four selected nodes per radial shell;
- three selected nodes per route/complement sector;
- direct/echo cell and value equality;
- C_{26} closure and common-sign relation;
- $72 + 72 + 12 = 156$ and lifted count 312;
- Subject-view spatial invariance and Carrier-view coherent evolution;
- every dressed station remaining on its existing Hopf leaf.

The central recommendation is therefore: construct the new family as a transposition of the verified affine event lattice, render the 13 exact stations as the primary meridian, and treat any smooth 3D curve as a labelled guide. That gives the requested space-like complement without adding an unproved metric or confusing translation fibers with Hopf/boost fibers.

ChatGPT can make mistakes. Check important info.